

Pilli, R. Saketh

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Cambridge Computer Science undergraduate focused on developing precise, accurate, and scalable software tools for science and engineering. Interested in the fields of computational science, formal methods, and algorithms.

EDUCATION

- Bach. (Hons) Computer Science** | *University of Cambridge* | cl.cam.ac.uk/teaching/2526/part1a.html Oct 2025 –
- **Ranked 3/41** on January examinations (Top Marks: Discrete Math, OOP, Mathematics)
 - **4th Place, Cambridge Graphics Competition:** Developed a “Lorentz Transform Visualizer” leveraging **GLSL** shaders to analytically trace and render relativistic effects through linear space-time transformations
 - **Undergraduate Academic Representative** for Computer Science and **Junior Member of Faculty Board**
- High School Diploma (High Honors)** | *BASIS Independent Fremont Upper School* Aug 2021 – Jun 2025
- **Relevant AP Scores:** 5/5 (Top Mark) Calculus BC, 5/5 Computer Science A (Java), 5/5 CS Principles, 5/5 Physics C Mechanics, 5/5 Physics C E&M, 5/5 Biology
 - **Higher Coursework:** Data Structures & Algorithms, Complex Analysis, Linear Algebra, Multivariable/Vector Calculus, Electrodynamics, Modern Cosmology
- MATH 170E+170S: Probability and Statistics** | *Summer Coursework: UCLA* Jun 2025 – Sep 2025
- MATH XM615: Real Analysis** | *Summer Coursework: Stanford University* Jun 2024 – Aug 2024

EXPERIENCE

- Software Engineering Intern** | *Hathority - AI.Data.Cloud* Aug 2024 – Dec 2024
- Developed collaboratively **ReleaseShield AI**, winning **5th place internationally** in “Boomi AI” competition, using the **Boomi** framework, AWS Sagemaker, ServiceNow, etc., using dependency mapping, developing the pre-update release management and risk assessment agentic AI: marketplace.boomi.com/solutions/hathority-releaseshield-ai-cda/
 - Developed an automatic video generator based on a prompt, leveraging **OpenAI APIs** and integrating through **Boomi Flow** for UI, **SnowflakeDB**, and **Groovy Script**. Also updated Boomi documentation for customers.
- Researcher** | *Aspiring Scholars Directed Research Program (ASDRP)* Mar 2024 – Sep 2024
- Led a team of 4 to computationally model crystallographic orientations of silicon surface roughness using **Quantum ESPRESSO** to simulate electron structures, optimizing microchip conductivity; performed data analysis with **pandas/matplotlib** and built 3D visualizations via **XCrysDen**
 - Presented findings at National ASDRP Blitz Talks and collaborated with laboratories to secure research funding
- Founder & President of Student Panel** | *BASIS Independent Fremont Upper School* Jan 2022 – Feb 2025
- Founded an inclusive Student Panel to bridge students, teachers, and parents; raised **\$25,000** and organized **10+** committees for student advocacy and volunteering events

HIGHLIGHTED RESEARCH AND PROJECTS

- The Theory, Properties, and Applications of Elliptic Functions** | *Primary Author w/ Dr Anuj Chaudhri* Current
- Review study of elliptic functions, focused on periodicity, ODEs, and series representations
 - Analyzed **Weierstrass** and **Jacobi** elliptic functions, **modular transformations**, and **applications** to elliptic integrals, algebraic equations, and **elliptic curves**
- The Creation, Detection, and Effects of Gravitational Waves** | *Primary Author w/ Dr Anuj Chaudhri* Current
- Authoring a review paper on the theoretical foundations of gravitational waves, leveraging **Einstein’s field equations** and the **Minkowski metric** to derive wave propagation in the **Transverse-Traceless (TT)** gauge and model radiation power via the quadrupole formula
- Simulating Natural Selection through AI Agents** | *High Honors Project* Mar 2025 – May 2025
- Built a Python simulation using **neural network AI agents** evolved through a custom **evolutionary algorithm**
 - Analytically explored emergent behaviors and adaptation by modeling **agent-environment interactions** and generational learning in dynamic ecosystems

CERTIFICATIONS & TECHNICAL SKILLS

Languages: Python (numpy, pandas, matplotlib), Java, OCaml, SQL, Cypher, GLSL
Tools/Platforms: Boomi (API Dev/Flow), SnowflakeDB, Quantum ESPRESSO, L^AT_EX, Claude Code
Concepts: Data Structures & Algorithms, Scientific Modelling, Statistical & Mathematical Methods, Databases
Certifications: Oracle Certified Java SE 8 Programmer, OpenEDG Certified Associate Python Programmer, Boomi Professional Certified (API Dev & Flow), Oracle Certified Foundations Java Assoc.